RESEARCH ARTICLE

IV Tenecteplase Before Thrombectomy Compared With Thrombectomy Alone in Patients With Stroke Due to a Large Vessel Occlusion

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Neurology® 2026;106:e214702. doi:10.1212/WNL.0000000000214702

Abstract

Background and Objectives

The benefit of IV thrombolysis (IVT) with alteplase before endovascular thrombectomy (EVT) compared with EVT alone has been shown to be limited and time dependent. Data on tenecteplase, its recommended alternative, are limited. We aimed to assess the efficacy and safety of IVT with tenecteplase plus mechanical thrombectomy (TNK + EVT) compared with EVT in patients with large vessel occlusion stroke and determine whether its potential benefit decreases with treatment time.

Methods

We conducted a retrospective pooled analysis of 2 nationwide, real-world registries of patients with anterior circulation large vessel occlusion stroke within 4.5 hours of known symptom onset and with no contraindication to thrombolysis, treated with TNK + EVT (TETRIS) or EVT (ETIS). The efficacy outcome was the 3-month modified Rankin Scale (mRS) score, analyzed in ordinal and dichotomized (mRS score ≤2) approaches. We used propensity score–weighted logistic regression to assess associations between treatment groups and outcomes of interest.

Results

Among 1,890 patients who were analyzed (TNK + EVT: n = 798; EVT: n = 1,092; median age 73 years [interquartile range 61–82]; 49.6% women), the median expected onset-to-thrombolysis time was 146 minutes [interquartile range 119–180]. More than half of patients (n = 1,063; 56.2%) were admitted first to a primary stroke center. All baseline characteristics were balanced between treatment groups after overlap weighting. Overall, TNK + EVT was associated with better 3-month functional outcome over the full mRS (weighted common odds ratio [OR] 1.53 [95% CI 1.29–1.82]; $p < 0.001$) and regarding functional independence (propensity score overlap weighting [PSOW]-OR 1.50 [95% CI 1.23–1.84]; $p < 0.001$). This benefit did not differ statistically between patients admitted first to a primary or comprehensive stroke center (p-interaction = 0.12). There was no significant effect of the expected onset-to-thrombolysis time on the association between TNK + EVT and better functional outcome (p-interaction = 0.11). There were no significant differences in parenchymal hematoma

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Glossary

ASMD = absolute standardized mean difference; **CSC** = comprehensive stroke center; **ETIS** = Endovascular Treatment of Ischemic Stroke; **EVT** = endovascular thrombectomy; **ICA** = internal carotid artery; **IRIS** = Improving Reperfusion Strategies in Acute Ischaemic Stroke; **IVT** = IV thrombolysis; **LVO** = large vessel occlusion; **MCA** = middle cerebral artery; **mRS** = modified Rankin Scale; **mTICI** = modified Thrombolysis in Cerebral Infarction; **NIHSS** = NIH Stroke Scale; **OR** = odds ratio; **PSC** = primary stroke center; **PSOW** = propensity score overlap weighting; **sICH** = symptomatic intracranial hemorrhage; **TETRIS** = Tenecteplase Treatment in Ischemic Stroke; **TNK + EVT** = IV thrombolysis with tenecteplase plus EVT.

(PSOW-OR 1.29 [95% CI 0.94–1.79]; p = 0.12) and symptomatic intracerebral hemorrhage (PSOW-OR 1.13 [95% CI 0.69–1.86]; p = 0.61) rates.

Discussion

Among patients treated within 4.5 hours of symptom onset, TNK + EVT was associated with better functional outcome than EVT, without safety concerns. This benefit does not seem to be time dependent. These findings support the routine use of tenecteplase before EVT in the early time window.

Classification of Evidence

This study provides Class II evidence that, in patients with stroke due to anterior circulation large vessel occlusion, IV tenecteplase plus thrombectomy is associated with better functional outcomes at 3 months compared with thrombectomy alone.

Trials Registration Information

NCT03776877 (ETIS registry) and NCT05534360 (TETRIS registry).

Introduction

IV thrombolysis (IVT) followed by endovascular thrombectomy (EVT) is the recommended treatment for patients with large vessel occlusion acute ischemic stroke (LVO stroke) within 270 minutes from symptom onset.[1-3] However, over the past few years, the benefit of IVT followed by EVT (IVT + EVT), compared with EVT alone, has been questioned. Six randomized clinical trials have assessed the benefit of IVT + EVT, compared with EVT alone, among patients directly admitted to a comprehensive stroke center (CSC).[4-9] These trials produced diverging results. Two concluded the noninferiority of EVT alone,[4,5] while 4 others did not prove this.[6-9] Overall, the Improving Reperfusion Strategies in Acute Ischaemic Stroke (IRIS) individual patient-level data meta-analysis did not prove the noninferiority of EVT alone to IVT + EVT regarding disability at 90 days[10] and found that IVT + EVT was associated with a higher final reperfusion rate. In addition, an IRIS ancillary study reported that the benefit associated with IVT + EVT was time dependent and statistically significant only if the expected symptom onset-to-administration of IVT (onset-to-thrombolysis) time was within 140 minutes.[11] However, in IRIS, patients who received IVT almost exclusively received alteplase (97.8%) as a thrombolytic rather than tenecteplase, which has become the recommended alternative for all patients with acute ischemic stroke with a known onset.[12]

Tenecteplase is a genetic variant of alteplase, with an improved fibrin specificity and longer half-life. A study-level meta-analysis of 11 randomized trials suggests that tenecteplase may be superior to alteplase in terms of functional outcome.[13] For patients with LVO stroke, this may be due in part to improved pre-EVT recanalization rates. Indeed, the EXTEND-IA TNK trial found that the use of tenecteplase before EVT was associated with higher pre-EVT recanalization rates than the use of alteplase (22% vs 10%).[14] An observational study of 377 patients who were directly admitted to a CSC from the EXTEND-IA TNK (IVT with tenecteplase followed by EVT [TNK + EVT] group) and SWIFT-DIRECT (EVT alone group) trials concluded that TNK + EVT resulted in improvement on the modified Rankin Scale (mRS) shift analysis.[15] Recent results from the BRIDGE-TNK randomized trial suggest that TNK + EVT may be superior to EVT alone among patients with LVO stroke who are directly admitted to a CSC in terms of 3-month functional independence (mRS score ≤2).[16] Contrary to alteplase, there are conflicting data on whether the benefit of TNK + MT is time dependent. While observational data suggest that the benefit associated with TNK + EVT is stronger among patients with an expected onset-to-thrombolysis time under 140 minutes,[15] in the BRIDGE-TNK trial, the benefit associated with TNK + EVT seemed stronger among patients with an expected onset-to-thrombolysis time above 140 minutes.

In this study, we aimed to assess whether IV tenecteplase plus thrombectomy is associated with better functional outcomes at 3 months compared with thrombectomy alone in patients with stroke due to anterior circulation large vessel occlusion,

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using real-world data from 2 nationwide French registries, the Endovascular Treatment of Ischemic Stroke (ETIS, EVT alone group) and the Tenecteplase Treatment in Ischemic Stroke (TETRIS, TNK + EVT group). In addition, we explored whether the 3-month neurologic outcome varied according to the expected onset-to-thrombolysis time and the type of first admission stroke center (primary stroke center [PSC] or CSC).

Methods

Our study was conducted according to the Strengthening the Reporting of Observational Studies in Epidemiology reporting guideline for observational studies.

Study Design and Data Sources

We conducted a retrospective pooled analysis of 2 prospective French multicenter nationwide real-world registries: patients in the EVT alone group were extracted from ETIS,[17] a registry of consecutive patients with LVO stroke undergoing EVT, and patients in the TNK + EVT group were selected from TETRIS,[18] a registry of consecutive patients with LVO stroke receiving tenecteplase. We included patients treated from January 2015 to March 2024 in the EVT alone group (ETIS) and from January 2015 to December 2022 (before the shortage of tenecteplase in Europe) in the TNK + EVT group (TETRIS). We included adult patients who met the following criteria: (1) LVO stroke of the anterior circulation, defined as an occlusion of either the intracranial internal carotid artery (ICA) or the first (M1) or second (M2) middle cerebral artery (MCA) segment, with or without a tandem cervical ICA occlusion; (2) treatment with IVT with 0.25 mg/kg (maximum 25 mg) tenecteplase and intended for EVT (TETRIS), or intended for EVT alone in the absence of a contraindication to IVT (ETIS) except for infarct size; and (3) an expected onset-to-thrombolysis time (defined further) within 4.5 hours of known symptom onset. We did not exclude patients based on infarct size or clinical severity because patients with large ischemic core were also present in the TETRIS registry.[19] We excluded patients who (1) were taking anticoagulants before stroke, (2) had missing symptom onset-to-baseline brain imaging time, and (3) had an isolated cervical ICA occlusion.

Standard Protocol Approvals, Registrations, and Patient Consents

This study was approved by the Sorbonne University Research Ethics Committee (CER–2021–1053). In ETIS (NCT03776877), informed consent was obtained from patients or their relatives. In TETRIS (NCT05534360), patients were informed of their participation and offered the possibility to opt out, in compliance with French law.

Clinical Data

The following variables were extracted from both registries: age, sex, vascular risk factors (high blood pressure, diabetes mellitus, current smoking), prestroke medication (antiplatelet, anticoagulants, statin), prestroke mRS score, neurologic severity measured using the NIH Stroke Scale (NIHSS), systolic blood pressure and blood glucose level on admission, and key time points (baseline brain imaging, IVT, groin puncture, and recanalization). For participants in the EVT alone group, the expected onset-to-thrombolysis time was derived by adding each individual patient's symptom onset-to-baseline brain imaging time to the median baseline brain imaging-to-thrombolysis time in TETRIS, depending on the type of stroke center in which it was performed (23 minutes for a CSC or 30 minutes in a PSC).

Imaging Data

Patients underwent either brain MRI or CT angiography on admission (day 0) and day 1. In both registries, baseline brain imaging, day-1 follow-up imaging, and digital subtraction angiograms of included patients were locally assessed by experienced stroke neurologists or neuroradiologists unblinded to the type of reperfusion therapy. The following variables were collected: (1) intracranial occlusion site: ICA or M1 and M2 MCA; (2) presence of a tandem cervical ICA occlusion; and (3) final modified Thrombolysis in Cerebral Infarction (mTICI) scores. Hemorrhagic transformation was assessed on 24-hour brain imaging, according to the European Cooperative Acute Stroke Study radiologic classification.[20]

Outcome Measures

Our primary outcome was better functional outcome at 3 months after stroke, defined as an improvement of at least 1 point on the mRS, collected by neurologists during poststroke consultations or phone interviews. Secondary outcomes were 3-month functional independence (mRS score ≤2) and final favorable recanalization (mTICI score 2b–3). Safety outcomes were parenchymal hematoma and symptomatic intracranial hemorrhage (sICH) defined as any parenchymal hematoma associated with an increase of at least 4 points in the NIHSS score within 36 hours of reperfusion therapy (i.e., modified Safe Implementation of Thrombolysis in Stroke-Monitoring definition).[21]

Statistical Analysis

Qualitative variables were expressed as counts and percentages and quantitative variables as mean ± SD or median (interquartile range), as appropriate. For our main analysis, we used a propensity score overlap weighting (PSOW) approach,[22] after multiple imputations for missing data, to account for imbalance in potential confounders for the association between treatment and outcomes of interest. Missing data rates for each variable are given in eTable 1. Imputations were performed under the missing-at-random assumption. The fully conditional specification method was used because it allows the use of different distributions across variables. The imputation model included all variables listed in Table 1, as well as treatment (IVT or no IVT) and 3-month mRS scores. Indeed, it is best practice to include the analysis model outcome in the imputation model when imputing

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Table 1 Baseline Characteristics

Characteristic	All patients (n = 1,890)	TNK + EVT (n = 798)	EVT alone (n = 1,092)	ASMD before PSOW[a] ()
Age, y, median (IQR)	73 (61–82)	74 (61–83)	72 (61–81)	9
Men, n (%)	953 (50.4)	384 (48.1)	569 (52.1)	8
Vascular risk factors, n (%)				
Hypertension	1,137 (60.2)	475 (59.5)	662 (60.6)	5
Diabetes mellitus	317 (16.8)	130 (16.3)	187 (17.1)	3
Current smoking	386 (20.4)	173 (21.7)	213 (19.5)	4
Statin use, n (%)	532 (28.1)	210 (26.3)	322 (29.5)	9
Prestroke mRS score >2, n (%)	119 (6.3)	64 (8.0)	55 (5.0)	12
NIHSS score, median (IQR)	16 (10–20)	15 (10–20)	16 (11–21)	18
SBP, mm Hg, mean (SD)	147.1 (27.1)	144.9 (28.0)	148.9 (26.1)	15
Glycemia, mmol/L, mean (SD)	7.0 (2.9)	6.9 (2.9)	7.2 (3.0)	4
ASPECTS, median (IQR)	7 (6–9)	8 (6–9)	7 (5–8)	32
Intracranial occlusion site, n (%)				5
ICA	378 (20.0)	157 (19.7)	221 (20.2)	
M1 MCA	1,164 (61.6)	475 (59.5)	689 (63.1)	
M2 MCA	348 (18.4)	166 (20.8)	182 (16.7)	
Cervical ICA occlusion, n (%)	288 (15.2)	133 (16.7)	155 (14.2)	7
Admission in a PSC, n (%)	1,063 (56.2)	564 (70.7)	499 (45.7)	52
Expected OTT, min, median (IQR)	146 (119–180)	144 (120–175)	149 (118–186)	8

Abbreviations: ASMD = absolute standardized mean difference; ASPECTS = Alberta Stroke Program Early CT Score; EVT = endovascular thrombectomy; ICA = internal carotid artery; IQR = interquartile range; MCA = middle cerebral artery; mRS = modified Rankin Scale; NIHSS = NIH Stroke Scale; OTT = onset-to-thrombolysis time; PSC = primary stroke center; PSOW = propensity score overlap weighting; SBP = systolic blood pressure; TNK + EVT = IV thrombolysis with tenecteplase plus EVT.
[a] After PSOW, both treatment groups were exactly balanced (ASMD = 0%, in Methods) regarding all baseline variables without excluding patients.

missing values of the analysis model's covariates.[23] All baseline variables (Table 1) were included in the logistic model used to estimate the propensity score of each patient to be intended for TNK + EVT (as opposed to thrombectomy alone), considering potential multicollinearity. Balance of baseline characteristics between the 2 treatment groups was assessed before PSOW by calculation of absolute standardized mean differences (ASMDs). An ASMD ≤10% was interpreted as a negligible difference.[24] It is noteworthy that, for all included baseline variables, a mathematical property of PSOW is that it leads to an exact balance (i.e., ASMD = 0%) between the treatment groups without excluding any patients.[22]

Associations between treatment groups and outcomes of interest were assessed using propensity score–weighted ordinal or binary logistic regression models and expressed as weighted odds ratios (PSOW-ORs) and their 95% CIs. The PSOW-ORs from each imputed data set were combined using Rubin rules. Potential heterogeneity in treatment effect (p-interaction) depending on (1) expected onset-to-thrombolysis time and (2) whether the patient was admitted first to a PSC or a CSC was assessed by including a product term in the logistic models. We nonparametrically examined the possibly nonlinear relationship between expected onset-to-thrombolysis time and each outcome using restricted cubic splines with 3 knots, corresponding to the 5th, 50th, and 95th percentiles of the expected onset-to-thrombolysis time. Tests for nonlinearity used the likelihood ratio test, comparing the model with only the linear term to a model with the linear and the cubic spline terms.[25] Based on the results of these tests, we assumed a linear association between expected onset-to-thrombolysis time and outcomes of interest.

All tests were 2-sided, and the significance level was set at $p < 0.05$. Analyses were performed using SAS 9.4 (SAS Inc., Cary, NC). G.T. and G.G. had full access to all data in the study and take responsibility for its integrity and the data analysis.

Data Availability

The data that support the findings of this study are available from the corresponding author on signing a data sharing agreement.

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Results

Study Population

Over the study period, 1,170 patients with anterior circulation LVO stroke were treated with tenecteplase and intended for EVT in the TETRIS registry and 8,449 patients were intended for EVT alone in the ETIS registry. Among them, 1,890 patients met our inclusion criteria (Figure 1), 798 in the TNK + EVT group (TETRIS registry) and 1,092 in the EVT alone group (ETIS registry). Reported reasons for not receiving IVT in the EVT alone group are presented in eFigure 1. Baseline characteristics are summarized in Table 1. Compared with the EVT alone group, several meaningful differences (ASMD >10%) were observed before PSOW in the TNK + EVT group: they were less severe on admission, with lower NIHSS scores and higher ASPECTS; they were more frequently initially admitted to a PSC. After PSOW, both treatment groups were exactly balanced (ASMD = 0%, in Methods) regarding all baseline variables included in Table 1. Patients in the TNK + EVT group underwent less frequently EVT than those treated with EVT alone (85.5% vs 96.2%).

Efficacy and Safety Outcomes

Compared with EVT alone, TNK + EVT was associated with better 3-month functional outcome over the full range of the mRS (weighted common OR 1.53 [95% CI 1.29–1.82]; $p < 0.001$; Table 2 and Figure 2). It was also associated with a higher rate of 3-month functional independence (PSOW-OR 1.50 [95% CI 1.23–1.84]; $p < 0.001$). Compared with EVT alone, TNK + EVT was not associated with higher odds of favorable recanalization at the end of the endovascular procedure (PSOW-OR 1.04 [95% CI 0.78–1.39]; $p = 0.78$). TNK + EVT was not associated with an increased risk of parenchymal hematoma (PSOW-OR 1.29 [95% CI 0.94–1.79]; $p = 0.12$) or sICH (PSOW-OR 1.13 [95% CI 0.69–1.86]; $p = 0.61$).

Expected Onset-to-Thrombolysis Time Analysis

Before PSOW, both groups did not differ in expected onset-to-thrombolysis times. Patients in the EVT alone group had longer onset-to-puncture and onset-to-reperfusion times (Figure 3). TNK + EVT remained significantly associated with better 3-month functional outcome for expected onset-to-thrombolysis times up to 3 hours and 10 minutes, but there was no significant treatment effect modification across expected onset-to-thrombolysis time (p-interaction = 0.11; Figure 3). This was also the case for 3-month functional independence (p-interaction = 0.27).

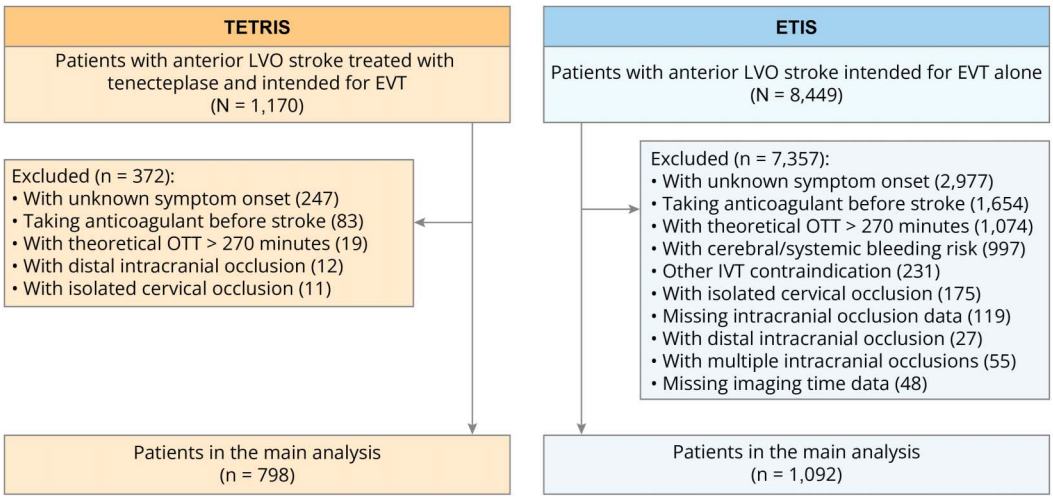
Subgroup Analysis

The PSOW-common OR for better functional outcome in patients first admitted to a PSC (n = 1,063) and in patients directly admitted to a CSC (n = 827) were 1.75 (95% CI 1.39–2.20) and 1.29 (95% CI 0.98–1.71), respectively (eFigure 2). There was no significant treatment effect modification between both treatment paradigms (p-interaction = 0.12).

Sensitivity Analysis

We performed a sensitivity analysis excluding patients with a large infarct core, defined as an ASPECTS ≤5 (n = 141 and n = 285 in the TNK + EVT and EVT-alone group, respectively), and those with a clearly established infarct in the EVT alone group (n = 96). In this analysis, TNK + EVT was associated with a better neurologic outcome (weighted common OR 1.59 [95% CI 1.31–1.92]; $p < 0.001$). There was no significant treatment effect modification across expected onset-to-thrombolysis time (p-interaction = 0.65; eFigure 3). TNK + EVT was associated with an increased risk of parenchymal hematoma (PSOW-OR 1.54 [95% CI 1.00–2.35]; $p = 0.048$) but not sICH (PSOW-OR 1.47 [95% CI 0.76–2.83]; $p = 0.25$).

Figure 1 Study Flowchart



EVT = endovascular thrombectomy; IVT = IV thrombolysis; LVO = large vessel occlusion; OTT = onset-to-thrombolysis time.

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Table 2 Primary and Secondary Outcomes

	TNK + EVT (n = 798)	EVT alone (n = 1,092)	Weighted odds ratio (95% CI)	*p* Value
Primary outcome[a]				
Lower 3-mo mRS score (ordinal analysis), median (IQR)	2 (1–4)	3 (1–5)	1.53 (1.29–1.82)	<0.001
Secondary outcomes				
3-mo mRS score ≤2, n (%)	407/798 (51.0)	441/1,092 (40.4)	1.50 (1.23–1.84)	<0.001
Final recanalization (mTICI 2b–3), n (%)	672/778 (86.4)	925/1,059 (87.4)	1.04 (0.78–1.39)	0.78
Parenchymal hematoma, n (%)	95/785 (12.1)	92/895 (10.3)	1.29 (0.94–1.79)	0.12
sICH (modified SITS-MOST), n (%)	35/785 (4.5)	40/895 (4.5)	1.13 (0.69–1.86)	0.61

Abbreviations: EVT = endovascular thrombectomy; mRS = modified Rankin Scale; mTICI = modified Thrombolysis in Cerebral Infarction; NIHSS = NIH Stroke Scale; sICH (modified SITS-MOST) = symptomatic intracerebral hemorrhage defined as an increase of 4 points or more in the NIHSS score either associated with a parenchymal hematoma (SITS-MOST); TNK + EVT = IV thrombolysis with tenecteplase plus EVT.
[a] For the mRS ordinal analysis, common odds ratio for better outcome (lower 3-month mRS score) were calculated.

Classification of Evidence

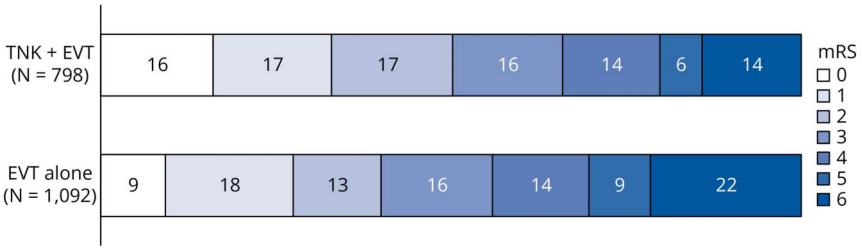
This study provides Class II evidence that, in patients with stroke due to anterior circulation large vessel occlusion, IV tenecteplase plus thrombectomy is associated with better functional outcomes at 3 months compared with thrombectomy alone.

Discussion

In this study, TNK + EVT was associated with better functional outcome than EVT alone in patients with anterior circulation LVO strokes treated within 4.5 hours of symptom onset. There was no significant treatment effect modification across expected onset-to-thrombolysis time and among patients admitted first either to a primary or a comprehensive stroke center. There were no significant differences in parenchymal hematoma or sICH between TNK + EVT and EVT alone.

Our main result should be interpreted in the continuing debate on the benefit of previous IVT before EVT. After the IVT bypass trials, which did not demonstrate the noninferiority of EVT alone among patients directly admitted to a CSC,[4-10] IVT + EVT remained the recommended treatment strategy for patients with LVO stroke.[26] However, in the IRIS meta-analysis of data from 2,313 individual patients, IVT + EVT was also not superior to EVT alone, with a nonsignificant trend in its favor (cOR 1.08 [95% CI 0.92–1.28]),[10] leaving unanswered the question of its benefit. This question has now shifted with the transition to tenecteplase for IVT in ischemic stroke. In this study, we present evidence that TNK + EVT is associated with a better functional outcome in routine clinical care. These findings are in line with those of a causal inference study, which found that TNK + EVT was associated with a better overall mRS score, but not with a higher functional independence rate (mRS score ≤2).[13] However, our study differs in several ways. First, we compared patients receiving routine clinical care, which is a less selective population than those included in randomized trials. Second, both treatment groups corresponded to the same healthcare system. Finally, all patients in the TNK + EVT group received the same dose (0.25 mg/kg) of tenecteplase. These findings also concur with the recent results of the BRIDGE-TNK trial, which found that TNK + EVT was associated with better 3-month functional independence rates (risk ratio 1.18 [95% CI 1.01–1.39]).[15] The differential effects of alteplase and tenecteplase before EVT may be primarily attributed to improved pre-EVT recanalization. Pre-EVT recanalization is associated with a better outcome[27,28] and occurs more frequently with

Figure 2 Three-Month mRS Scores After Propensity Score Overlap Weighting



Data represent the proportion of patients. EVT = endovascular thrombectomy; mRS = modified Rankin Scale; TNK + EVT = IV thrombolysis with tenecteplase plus EVT.

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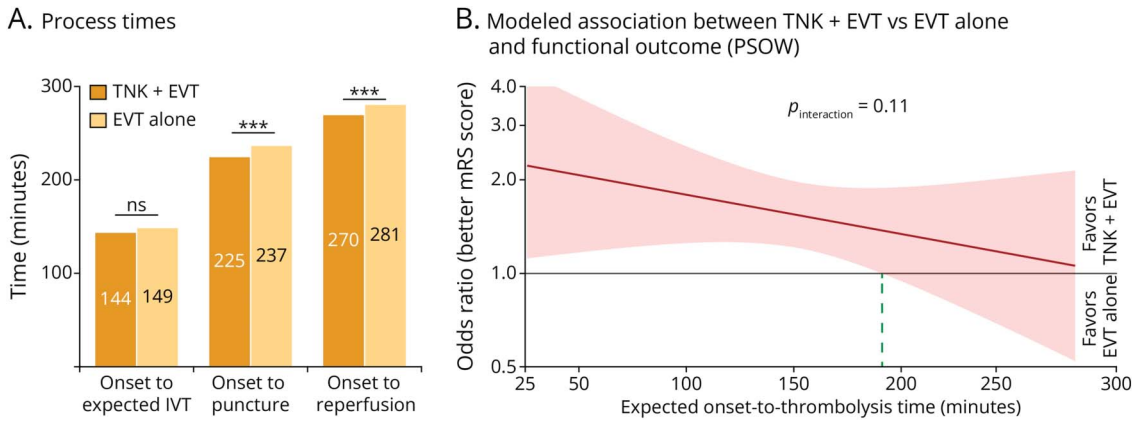
Figure 3 Process Times and Modeled Association of TNK + EVT vs EVT Alone and Functional Outcomes



In A, data represent the proportion of time. Statistical analysis: ***$p < 0.001$; n.s. = nonsignificant. In B, the red line indicates best model fit of the log odds ratio for a favorable shift in the 3-month mRS scores associated with tenecteplase plus endovascular thrombectomy (TNK + EVT) vs endovascular thrombectomy (EVT) alone and the expected onset-to-thrombolysis time. The shaded area reflects the 95% CI for the adjusted common odds ratio. An adjusted common odds ratio greater than 1 indicates better outcomes (favorable shifts in the mRS scores) associated with TNK + EVT. The dashed line indicates the time at which the lower bound of the 95% CI crossed 1.0.

tenecteplase, as demonstrated by the EXTEND-IA TNK trial.[14] Although we did not specifically assess pre-EVT recanalization in our cohort, the lower rate of EVT in the TNK + EVT group (85.5% vs 96.2%) likely reflects this. Another hypothesis could be the microvascular effect of IVT, supported by 2 positive intra-arterial thrombolysis trials after successful recanalization.[29,30] However, there are no comparative data on the microvascular effects of alteplase and tenecteplase. Altogether, pending the results of the still ongoing DIRECT-TNK (NCT 05199194), these results strengthen the rational for the use of TNK + EVT in patients with LVO stroke.

Our results also contribute to another question raised in the wake of the IVT bypass trials regarding the relationship between the benefit of previous IVT before EVT and the expected onset-to-thrombolysis time. While it has been well demonstrated that the separate benefits of IVT and EVT decrease, respectively, with the onset-to-thrombolysis and onset-to-reperfusion times,[31,32] current data on the benefit of IVT before EVT differ depending on the thrombolytic. On one hand, it has been demonstrated by an ancillary analysis of IRIS that the benefit of previous IVT with alteplase before EVT decreases with the expected onset-to-thrombolysis time.[11] On the other hand, the picture is less clear for tenecteplase. An observational study found that TNK + EVT was associated with better outcomes only for patients treated within 140 minutes,[15] whereas the preplanned subgroup analysis of the BRIDGE-TNK trial suggested the opposite.[16] In our study, we did not observe a significant effect of the expected onset-to-thrombolysis time on the benefit of previous IVT with tenecteplase within 4.5 hours of symptom onset. Although we cannot rule out the possibility of insufficient statistical power, this finding may also result from the intrinsic superiority of tenecteplase over alteplase before EVT in patients with LVO stroke, as suggested by the EXTEND-IA TNK trial.[14] Pooled individual patient data analysis of the BRIDGE-TNK and DIRECT-TNK trials will be of high interest to further investigate this relationship.

Finally, our study stands apart from available data comparing TNK + EVT and EVT alone as it provides insight into the benefit of TNK + TM not only among patients directly admitted to a CSC but also those admitted first to a PSC, who were excluded from the IVT bypass trials. In this study, although there was a numerically higher treatment effect size among patients admitted first to a PSC, there was no significant interaction between the type of first admission stroke center and the benefit associated with TNK + EVT. This result was surprising because several observational studies that focused on patients first admitted to a PSC reported that IVT with alteplase was associated with higher rates or early recanalization and better functional outcome,[33-38] which could have led to a higher benefit of TNK + EVT in this setting. Given that a specific emphasis was made in the TETRIS registry to include all patients intended for EVT, including those who improved or recanalized and did not undergo the procedure, it is unlikely that we underestimated the benefit of TNK + EVT in this setting. However, this could be explained by a higher benefit of tenecteplase compared with alteplase before EVT in CSCs. Our results support the need to widen the use of tenecteplase among all patients with LVO stroke, independently of the treatment paradigm.

Our study has several strengths. It is based on 2 large multicentric LVO stroke patient registries with similar practices. We used a modern statistical approach, PSOW, which allows an optimal control of measured confounders without exclusion of patients. However, our study also has several limitations. It was based on retrospectively collected data, with unmeasured confounding factors that could not be accounted

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for by PSOW and that may have influenced our results. Because the 2 treatment groups correspond to very different patient populations before adjustment, residual confounding differences may persist after PSOW. Our main outcome (3-month mRS score) was assessed unblinded and in real-world conditions, which may represent a potential misclassification bias. However, differential misclassification bias seems unlikely because the data were collected prospectively outside the scope of this or any other comparative study. In our data set, the mRS score at hospital discharge was not routinely collected, precluding its use for imputing missing 3-month mRS values. The baseline and follow-up brain imaging were not centrally assessed but read locally by experienced neurologists or neuroradiologists. Patients were included over a long period (9 years), and changes in clinical practice over time may have introduced bias. Regarding the declared reason for not receiving IVT in ETIS, we cannot exclude that other nonreported factors influenced this decision. Reliable pre-EVT recanalization data were unavailable for most patients, limiting mechanistic insight into the benefit of TNK + EVT. Patients in the TNK + EVT group had a longer onset-to-reperfusion time, which may have influenced their outcome. Finally, although all patients were from a single healthcare system (France), both treatment groups were mostly drawn from different stroke centers.

IVT with tenecteplase before EVT was associated with better functional outcome than EVT alone in patients with anterior circulation LVO strokes treated within 4.5 hours of symptom onset, without safety concerns. There was no significant interaction between the onset-to-thrombolysis time and the type of first admission stroke center on the treatment effect of tenecteplase use before EVT in this early time window. These findings support the routine tenecteplase use before EVT in the early time window.

Author Contributions

G. Gerschenfeld: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. B. Lapergue: drafting/revision of the manuscript for content, including medical writing for content; study concept or design. P. Seners: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. M. Piotin: major role in the acquisition of data. N. Chausson: major role in the acquisition of data. L. Scarcia: major role in the acquisition of data. C. Arquizan: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. V. Costalat: major role in the acquisition of data. S. Olindo: major role in the acquisition of data. G. Marnat: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. B. Guillon: major role in the acquisition of data. R. Bourcier: major role in the acquisition of data. M. Yger: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. J. Allard: major role in the acquisition of data. S. Richard: major role in the acquisition of data. B. Gory: major role in the acquisition of data. F. Pico: major role in the acquisition of data. F. Macian Montoro: major role in the acquisition of data. C. Mounayer: major role in the acquisition of data. W. Ben Hassen: major role in the acquisition of data. L. Venditti: major role in the acquisition of data. J. Caroff: major role in the acquisition of data. F. Clarençon: major role in the acquisition of data. A. Consoli: major role in the acquisition of data. G. Turc: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. S. Alamowitch: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data.

Study Funding

This research has received funding from the DMU Neurosciences of the APHP, Sorbonne Université, and the "Investissements d'avenir" ANR-10-IAIHU-06. The TETRIS registry received funding from Boehringer Ingelheim.

Disclosure

G. Gerschenfeld reports consulting fees from Truffle Capital and funding from the Programme Hospitalier de Recherche Clinique, the DMU Neurosciences of APHP, Sorbonne Université, and the "Investissements d'avenir" ANR-10-IAIHU-06. B Lapergue reports grant from Balt USA, MicroVention, and Phenox. P. Seners reports consulting fees from Boehringer Ingelheim. C. Arquizan reports lecturing fees from Amgen, Novartis, and Medtronic Vascular and consulting fees from Medtronic Vascular. V. Costalat reports consulting fees from Balt USA, Johnson & Johnson Health Care Systems, Medtronic USA, MicroVention, Penumbra, and Stryker Corporation. G. Marnat reports consulting fees from Stryker Neurovascular, Microvention Europe (Terumo Neuro), Balt Extrusion, Sim, and Cure; and paid lectures from Medtronic, Wallaby Phenox, Bracco, Penumbra, and Johnson & Johnson. M. Yger reports lecturing fees from Lundbeck. F. Clarençon reports consulting fees from Medtronic, Balt Extrusion, Wallaby Phenox, Microvention, ClinSearch, Penumbra, and Stryker; and is a board member of Artedrone. G. Turc reports lecturing fees from Guerbet France and consulting fees for Neurologica and AI-Stroke. S. Alamowitch reports consulting fees from Amgen, AstraZeneca, Boehringer Ingelheim, and Pfizer Pharma GMBH; and funding from the Programme Hospitalier de Recherche Clinique, the DMU Neurosciences of APHP, Sorbonne Université, the "Investissements d'avenir" ANR-10-IAIHU-06 and Boehringer Ingelheim. All other authors report no disclosures relevant to the manuscript. Go to Neurology.org/N for full disclosures.

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Publication History

Received by *Neurology*® July 29, 2025. Accepted in final form December 17, 2025. Submitted and externally peer reviewed. The handling editor was Editor-in-Chief José Merino, MD, MPhil, FAAN.

Appendix Coinvestigators

Coinvestigators are listed at Neurology.org/N.

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